University of Toronto Mississauga

Mathematical and Computational Sciences

MAT102H5S - Make Up Test

Test Booklet

Duration - 110 minutes

No Aids Permitted

Surname/Family Name: ______________________________

First/Given Name: ______________________________

Student Number: ______________________________

UTORid: ______________________________

Email: ______________________________

This exam contains 6 pages (including this cover page) and 5 problems. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 5 (page 6). No aids are permitted on this examination. Examples of illegal aids include, but are not limited to textbooks, notes, calculators, or any electronic device.

Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 6). When filling out the scantron, ensure that bubbles are completely filled and make complete erasures. Any failure to follow these instructions which results in the scanner not being able to read your work will result in a 3 mark deduction. You will not be penalized for incorrect answers.

Your solutions to Problems 1 - 4 must be written in the space provided. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this test.**

Unless otherwise indicated, you are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work** in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work, will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- You may use any result mentioned explicitly in the readings, assignments, or in-class worksheets without proof. To use a result which is not covered by the aforementioned sources, you must provide a proof of the result as part of your work.

The Multiple Choice Booklet is printed one-sided, and may thus be used for rough work.

1. For any two subsets A and B of $\mathbb{Z}$, we define their symmetric difference as $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

   (i) (3 points) Show that if A, B, and C are sets, then $A \Delta C \subseteq (A \Delta B) \cup (B \Delta C)$.

   (ii) (3 points) Define a relation on $\mathcal{P}(\mathbb{Z})$ by saying that $A \sim B$ if and only if $|A \Delta B|$ is finite. Show that $\sim$ is an equivalence relation.

2. Recall that $B^A = \{f : A \to B\}$. If A, B, and C, are non-empty sets, define the map $E : B^A \times A \to B$ given by $E(f, x) = f(x)$.

   (i) (1 point) Show that E is always surjective.

   (ii) (1 point) Show that if $|A| > 1$ then E is not injective.

   (iii) (3 points) Show that if $|A| = 1$ then E is injective.

3. Let p_n denote the nth prime, in the usual order.

(i) (3 points) Argue that $p_{n+1} \leq p_1 \cdots p_n + 1$ for all $n \in \mathbb{Z}^+$.

(ii) (3 points) If $r \neq 0, 1$ is a real number, show that $\displaystyle\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}$ for all $n \in \mathbb{Z}^+$.

(iii) (3 points) Show that $p_n \leq 2^{2^{n-1}}$ for all $n \in \mathbb{Z}^+$.

4. (3 points) Suppose that $a, b, n \in \mathbb{Z}^+$ and that $a \equiv b \pmod{n}$. Show that $\gcd(a, n) = \gcd(b, n)$.